

Water hammer in domestic hot water pipes

The physics of pressure surges, and how the waterhammer model decides whether a pipe needs protection

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When a tap or an appliance valve shuts, the water flowing in the pipe behind it has to stop. Its momentum does not vanish: it becomes a pressure wave that runs back and forth along the pipe, and the pressure at the valve can rise by many times the normal supply pressure. This is *water hammer*. This note sets out the physics the waterhammer model uses to decide whether a domestic hot water branch needs protection against it: how fast the wave travels, how large the surge is for fast and slow valves, how the model simulates the transient, when the water column can break apart, and how large a gas-cushion arrester has to be. It ends with the acceptance limits, a worked example and screening results. The Python package and the Excel workbook implement exactly these equations, and agree with each other to machine precision.

The setting

Consider a straight, horizontal pipe of length L (m) with a circular bore of inside diameter D (m) and a wall of thickness e (m). The bore's cross-sectional area is $A = \pi D^2/4$ (m²). One end of the pipe joins a point where the pressure stays nearly constant whatever happens in the pipe: a much larger main, a manifold, a hot water cylinder or an expansion vessel. Call this the *reflection point*. The other end has a valve: a tap, a shower valve or an appliance's solenoid valve.

Water of density ρ (kg/m³) flows through the pipe at a volume flow rate Q (m³/s), so its mean velocity is $v = Q/A$ (m/s).¹ With no flow, the gauge pressure everywhere in the pipe is the *rest pressure* p_s (Pa), the pressure the supply holds at the reflection point. In this note a gauge pressure is measured relative to the atmosphere, and 1 bar = 10⁵ Pa.

The setting is a domestic hot water branch: flows of 1 to 6 L/min, pipes of about 10 to 15 mm outside diameter in copper, PE-X (cross-linked polyethylene), PB (polybutylene) or multilayer composite (MLCP), rest pressures of 2.5 to 4 bar and water at 50 to 60 °C.

¹ At 6 L/min, $Q = 10^{-4}$ m³/s. In 10 mm copper (bore 8.8 mm, $A = 6.1 \times 10^{-5}$ m²) that is $v = 1.64$ m/s.

The mechanism

When the valve shuts, the layer of water touching it stops first. Its pressure rises, which compresses that water slightly and stretches the pipe wall around it. The next layer upstream then meets stationary water and stops in turn. A front of raised pressure, with stationary water behind it, so runs upstream at the *wave speed* a (m/s). As the section on wave speed shows, a is close to the speed of sound in water for copper and much lower for plastic.

The front reaches the reflection point after a time L/a . At that moment all the water in the pipe is at rest and above the rest pressure, while the reflection point is still at p_s . Water therefore starts to flow back out of the pipe, and a wave that restores the pressure to p_s travels down to the valve. It arrives at time

$$T_c = \frac{2L}{a}, \quad (1)$$

where T_c (s) is the *critical time*.² Because the water is now moving away from the closed valve, the pressure there falls below p_s , and a low-pressure wave goes back upstream. With no friction the cycle would repeat for ever, the pressure at the valve swinging above and below p_s with period $4L/a$. Friction damps it, and in real installations so do trapped air, flexible hoses and fittings that move.

² A 10 m copper run has $T_c \approx 15$ ms; the same run in PE-X has $T_c \approx 70$ ms.

Surge from a fast closure

Let Δv (m/s) be the drop in velocity across the wave front; for a valve that shuts completely, $\Delta v = v$. Let Δp (Pa) be the rise in pressure across the front.

Take a short length of pipe around the front. Relative to the pipe, the front moves upstream at $a - v$, while the water ahead of it moves downstream at v , so the front meets new water at the relative speed a . In a short time δt (s) it therefore brings a mass $\rho A a \delta t$ (kg) of water to a stop (strictly, from v to $v - \Delta v$). Newton's second law says that the impulse of the net force equals the change of momentum. The net force is the pressure difference Δp acting on the area A , so

$$\Delta p A \delta t = (\rho A a \delta t) \Delta v \implies \Delta p = \rho a \Delta v. \quad (2)$$

This is Joukowsky's equation.³ It gives the full surge whenever the valve stops the flow before the first returning wave can reach it, that is when the valve's effective closure time t_v (s) is no more than T_c . The *effective* closure time is the time over which the valve actually throttles the flow, usually only the last part of the handle's travel. The surge depends neither on the length of the pipe nor on the pressure already in it.⁴

³ Joukowsky (1904).

Wave speed

The wave speed follows from conservation of mass across the front. Water keeps arriving at the front at velocity v but leaves it at $v - \Delta v$, so in time δt a volume $A \Delta v \delta t$ of extra water has to be stored in the length $a \delta t$ the front has swept. There are two places to store it. The water is compressed: its density rises by a fraction $\Delta p/K$, where K (Pa) is the water's *bulk modulus*. And the pipe swells: its bore area rises by a fraction $\Delta A/A$. So

$$A \Delta v \delta t = A a \delta t \left(\frac{\Delta p}{K} + \frac{\Delta A}{A} \right). \quad (3)$$

⁴ In copper, $\rho a \approx 13$ bar per m/s. A flow of 1 m/s stopped instantly gives 13 bar, more than six times the 2 bar limit set out under Acceptance limits.

How much the pipe swells. The pressure rise Δp adds a hoop (circumferential) stress $\Delta p D/(2e)$ (Pa) to a thin wall. If E (Pa) is the wall material's Young's modulus, the hoop strain, the fractional increase in diameter, is $\Delta p D/(2eE)$, and the area grows by twice that fraction: $\Delta A/A = \Delta p D/(eE)$. Real pipes are not quite thin, and how the pipe is held lengthwise changes the answer slightly, because a pipe that cannot lengthen develops an axial stress as it swells. Both effects are carried by a dimensionless *restraint factor* ψ , so that

$$\frac{\Delta A}{A} = \psi \frac{\Delta p D}{E e}. \quad (4)$$

For a thick-walled pipe whose material has Poisson's ratio ν (dimensionless: the lateral contraction per unit of stretch),⁵

$$\psi = 2 \frac{e}{D} (1 + \nu) + \frac{D}{D + e} \phi, \quad (5)$$

where $\phi = 1 - \nu^2$ for a pipe anchored against lengthwise movement throughout (clipped at short intervals, or buried in a screed), $\phi = 1$ for a pipe free to move lengthwise, and $\phi = 1 - \nu/2$ for a pipe anchored only at its upstream end.⁶

The result. Substituting $\Delta v = \Delta p/(\rho a)$ from Equation 2 into Equation 3 and cancelling $A \delta t \Delta p$ gives $1/(\rho a) = a(1/K + \psi D/(Ee))$, which rearranges to the Korteweg formula

$$a = \sqrt{\frac{K/\rho}{1 + \psi \frac{KD}{Ee}}}. \quad (6)$$

In a rigid pipe ($E \rightarrow \infty$), a becomes $\sqrt{K/\rho}$, the speed of sound in water: about 1500 m/s. Copper ($E \approx 117$ GPa) is stiff enough to stay close to that. Plastics are more than a hundred times more flexible, so the pipe term dominates and a falls to a few hundred metres per second (Figure 1).

Properties used. For water, $K = \rho c^2$, where c (m/s) is the speed of sound in water; ρ and c come from the IAPWS-95 formulation as functions of temperature.⁷ K is largest near 50 °C, so hot water gives a slightly *higher* surge than cold water in a copper pipe. For plastics, E has to be the short-term (dynamic) modulus, because a surge loads the wall for milliseconds, not years. It falls steeply with temperature: for PE-X the model uses 0.70 GPa at 20 °C and 0.42 GPa at 60 °C. These are indicative values, and a manufacturer's figure should replace them where one exists.

Slower closure

If the valve takes longer than T_c to shut, waves returning from the reflection point arrive while it is still closing and cancel part of the rise.

⁵ Wylie and Streeter (1993), §2.4.

⁶ As $e/D \rightarrow 0$ the first term vanishes and $D/(D + e) \rightarrow 1$, so a thin pipe free to move has $\psi = 1$, the textbook form.

⁷ Wagner and Pruiß (2002).

Plastic pipe carries pressure waves four to five times slower

Wave speed at 55 °C, pipe anchored against axial movement

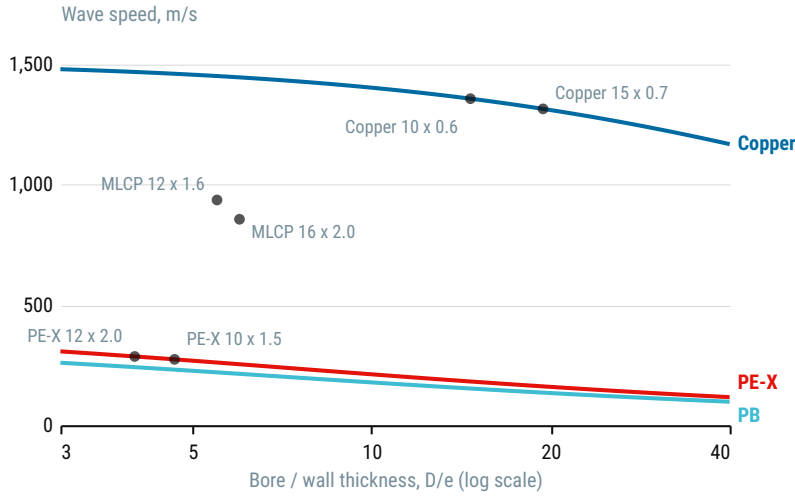


Figure 1: Wave speed a against the ratio of bore to wall thickness, D/e , at 55 °C. The multilayer (MLCP) points assume an aluminium layer 0.2 mm thick; the wall's hoop stiffness is the thickness-weighted mean of the aluminium and PE-RT (the plastic either side of it), because all the layers share one hoop strain.

Michaud's formula. Suppose the velocity at the valve falls at a steady rate, from v to zero over t_v . Each instant of closure sends a small Joukowski step up the pipe, so the pressure at the valve climbs in proportion to the velocity lost so far, $\rho a v t / t_v$ at time t (s) after closure starts. At $t = T_c$ the first returning wave arrives and the climb stops: from then on the steps still being sent are cancelled by those coming back. The peak is therefore the Joukowski surge scaled by T_c / t_v :

$$\Delta p_{\max} = \rho a \Delta v \frac{T_c}{t_v} = \frac{2\rho L \Delta v}{t_v} \quad (t_v > T_c). \quad (7)$$

The wave speed has cancelled. For slow closures the surge depends only on the run length, the velocity and the closure time, not on the pipe material.⁸

Real valves. A tap discharging to atmosphere does not reduce the velocity at a steady rate. Its flow obeys the orifice law

$$Q = \tau(t) Q_0 \sqrt{H_v / H_{v0}}, \quad (8)$$

where τ (dimensionless) is the fraction of the valve's opening that is still open, Q_0 (m³/s) is the steady flow before closure, and H_v (m) is the gauge pressure just upstream of the valve expressed as a height of water, $H_v = p / (\rho g)$, with p (Pa) that gauge pressure and $g = 9.81 \text{ m/s}^2$ the acceleration due to gravity. H_{v0} is its steady value while the water flows. With τ falling at a steady rate, most of the flow is choked only near the end of the stroke, which is why the effective closure time is shorter than the handle's travel. When $t_v \gg T_c$ the water behaves as a rigid column, and Newton's law for the whole column gives

$$\frac{L}{gA} \frac{dQ}{dt} = H_R - h_f(Q) - H_v, \quad (9)$$

⁸ For a steady fall in velocity in a frictionless pipe, Equation 7 is the exact peak. The pressure then oscillates between p_s and this peak, about a mean of $\rho L \Delta v / t_v$, which is what treating the water as a rigid column predicts.

where $H_R = p_s/(\rho g)$ (m) is the head at the reflection point and h_f (m) the head lost to friction along the pipe at flow Q . The model uses this limit to check its simulation.

Simulating the transient

To handle any valve and closure time, the model solves the one-dimensional water hammer equations.⁹ Let x (m) be the distance along the pipe from the reflection point, t (s) time, and $H(x, t)$ (m) and $Q(x, t)$ (m³/s) the local pressure head and flow. Mass and momentum conservation give

⁹ Wylie and Streeter (1993), ch. 3.

$$\frac{\partial H}{\partial t} + \frac{a^2}{gA} \frac{\partial Q}{\partial x} = 0, \quad (10)$$

$$\frac{\partial Q}{\partial t} + gA \frac{\partial H}{\partial x} + \frac{f Q |Q|}{2DA} = 0, \quad (11)$$

where f (dimensionless) is the Darcy friction factor. The model takes $f = 64/Re$ for laminar flow ($Re < 2300$) and the Swamee–Jain formula $f = 0.25 / [\log_{10}(\epsilon/3.7D + 5.74/Re^{0.9})]^2$ above that. Here $Re = \rho v D / \mu$ is the Reynolds number (dimensionless), μ (Pa s) the water's dynamic viscosity and ϵ (m) the roughness of the bore. The friction head over the whole pipe is then $h_f = f (L/D) v^2 / (2g)$.

The method of characteristics. Along lines in the (x, t) plane whose slope is $dx/dt = \pm a$, the two partial differential equations reduce to ordinary ones. Divide the pipe into N equal reaches of length $\Delta x = L/N$ (m), and step time by $\Delta t = \Delta x/a$ (s), so that the characteristic lines join neighbouring grid points. Write H_i and Q_i for the values at node i (node 0 is the reflection point, node N the valve) at the previous time step, and H_P, Q_P for the new values at node i . Then

$$C^+ : H_P = C_P - B Q_P, \quad C_P = H_{i-1} + B Q_{i-1} - R Q_{i-1} |Q_{i-1}|, \quad (12)$$

$$C^- : H_P = C_M + B Q_P, \quad C_M = H_{i+1} - B Q_{i+1} + R Q_{i+1} |Q_{i+1}|, \quad (13)$$

where $B = a/(gA)$ (s/m²) is the pipe's characteristic impedance and $R = f \Delta x / (2gDA^2)$ (s²/m⁵) its friction coefficient per reach. Solving the pair gives, at each interior node, $H_P = (C_P + C_M)/2$ and $Q_P = (C_P - C_M)/(2B)$.

Ends of the pipe. At the reflection point the head is fixed, $H_P = H_R$, so the C^- equation gives $Q_P = (H_R - C_M)/B$. At the valve, the C^+ equation and the orifice law (Equation 8) together give a quadratic in Q_P , whose positive root is

$$Q_P = \frac{1}{2} \left(-kB + \sqrt{k^2 B^2 + 4k C_P} \right), \quad k = \frac{(\tau Q_0)^2}{H_{v0}} \text{ (m}^5/\text{s}^2\text{)}, \quad (14)$$

with $Q_P = 0$ once the valve is shut. The open fraction falls at a steady rate, $\tau = 1 - t/t_v$. The simulation starts from steady flow, with

the head falling linearly from H_R at the reflection point to $H_{v0} = H_R - h_f$ at the valve, and runs until six periods of $2L/a$ after the valve has shut. The tests check that it reproduces Joukowski's surge for an instantaneous closure (within 3%) and the rigid-column result of Equation 9 for slow closures (within 5%).

Only valves that close within $2L/a$ see the full Joukowski surge

Peak surge at the valve; orifice valve, linear area closure, 3 bar rest pressure

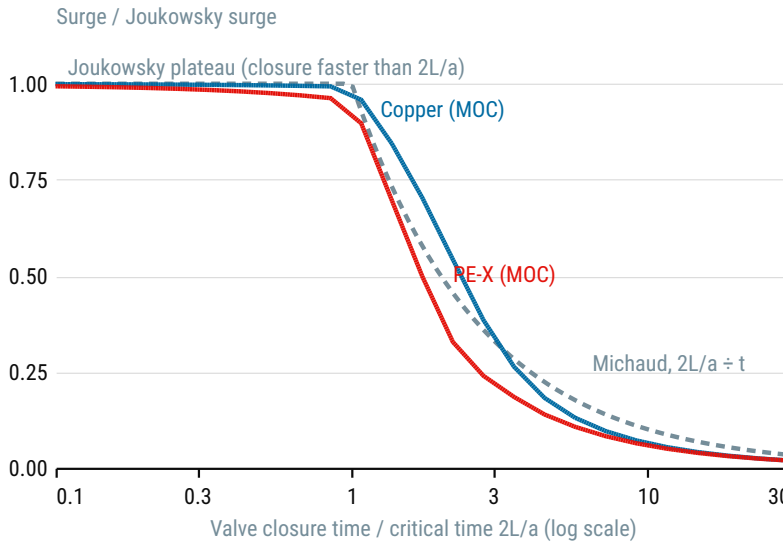
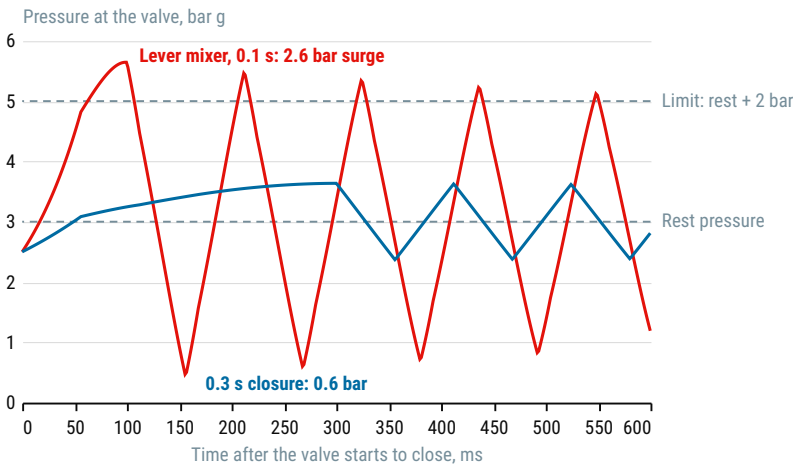


Figure 2: Peak surge as a fraction of Joukowski's ρav , against the valve's closure time measured in units of the critical time $2L/a$. The dashed line is Equations 2 and 7. The solid lines are the simulation with an orifice valve whose opening falls at a steady rate.

Figure 2 shows the outcome. Below $t_v \approx T_c$ every closure gives the full Joukowski surge. Above it the surge falls roughly as $1/t_v$. The simulated curve can lie above or below Michaud's line, depending on how hard the orifice throttles the flow compared with the rest pressure.¹⁰ The model therefore takes the *larger* of the formula and the simulation as the design surge, written Δp_d (Pa).

Tripling the closure time brings the surge within the limit

PE-X 12 x 2.0, 8 m run, 6 L/min, 55 °C (method of characteristics)



¹⁰ Allievi's constant, $av/(2gH_{v0})$ (dimensionless), measures this: the ratio of the Joukowski head rise to twice the head at the valve. In the two examples Figure 3, it is 0.1 and 0.3. The red line starts to close. Each spike is the wave returning from the reflection point, at intervals of $4L/a$. Friction damps the oscillation only slowly.

Column separation

Each high-pressure wave is followed by a low-pressure one of about the same size (Figure 3). If the absolute pressure anywhere falls to the water's vapour pressure p_{vap} (Pa), the pressure at which water boils at its current temperature, the column breaks and a vapour cavity forms. When the columns rejoin, the impact can exceed Joukowsky's surge.¹¹ The model detects this but does not simulate it: any minimum pressure below p_{vap} counts as a failure.¹²

¹¹ Wylie and Streeter (1993), ch. 8.

¹² At 55 °C, $p_{vap} = 0.16$ bar absolute, or -0.85 bar gauge. With a rest pressure of 3 bar the column separates once the surge exceeds about 3.9 bar.

Gas-cushion arresters

A water hammer arrester puts a volume of gas, kept apart from the water by a piston or membrane, next to the valve. To a first approximation it must absorb the kinetic energy of the moving column, $\frac{1}{2}\rho ALv^2$ (J), while the pressure rises by no more than an allowed amount Δp_a (Pa). Let p_0 (Pa, absolute) be the gas's precharge pressure and V_0 (m³) its volume at that pressure, and suppose the gas is compressed polytropically, so that pV^n stays constant with index n (dimensionless; $n = 1$ for slow, isothermal compression and $n = 1.4$ for fast, adiabatic compression of air or nitrogen). The work done on the gas in compressing it from p_0 to $p_0 + \Delta p_a$ then equals the kinetic energy:

$$\frac{1}{2}\rho ALv^2 = \frac{p_0 V_0}{n-1} \left[\left(\frac{p_0 + \Delta p_a}{p_0} \right)^{(n-1)/n} - 1 \right], \quad (15)$$

which tends to $p_0 V_0 \ln[(p_0 + \Delta p_a)/p_0]$ as $n \rightarrow 1$. Solving for V_0 gives the precharge volume. The model takes $n = 1.4$, which needs more gas than $n = 1$, and sets p_0 to the rest pressure plus atmospheric pressure. It ignores the energy taken up by the stretching pipe and the compressed water, which makes the estimate conservative. The result shows the scale needed; the product itself should be chosen from the manufacturer's data, or by PDI-WH 201 in US practice.¹³

¹³ Plumbing and Drainage Institute (2010).

Acceptance limits

Check	Default	Source
Surge above rest pressure, Δp_d	at most 2 bar	Technical rules for drinking water installations (EN 806-2 with DIN 1988-200), as quoted in VDI 6006, §4.3
Peak pressure, $p_s + \Delta p_d$	at most the pipe and fitting rating at the water temperature (10 bar by default)	Water Supply (Water Fittings) Regulations 1999, Schedule 2, paragraphs 3, 5 and 12
Minimum pressure	above p_{vap}	Column separation
Velocity, v (advisory)	at most 1.5 m/s for hot water	Uniform Plumbing Code 610.12; DIN 1988-300 allows 2 m/s in single connection pipes

UK and European practice sets a performance requirement: keep the pressure rise caused by operating a valve within 2 bar (BSI, 2005;

Table 1. The checks the model applies. Failing any of the first three means mitigation is required. Every limit is an input to the model.

DIN, 2012; VDI, 2017). The UK Water Fittings Regulations also require fittings to resist pressure surges and to withstand 1.5 times the maximum pressure they are designed to operate at (UK Government, 1999). US codes are prescriptive instead: the International Plumbing Code and the Uniform Plumbing Code require an arrester certified to ASSE 1010 wherever quick-closing valves are used (ICC, 2021; IAPMO, 2021; ASSE, 2004). Table 1 lists the checks.

When mitigation is needed the model offers three remedies. The first is an arrester at the valve, sized by Equation 15 with Δp_a the smaller of the surge limit and the margin below the pipe rating. The second is the shortest effective closure time that meets the limit. The Python model finds it by bisection on the larger of Michaud's formula and the simulation; the Excel workbook gives Michaud's estimate, $t_v = 2\rho Lv/\Delta p_a$. The third is the largest flow at which even an instantaneous closure passes, from Equation 2: $v \leq \Delta p_a/(\rho a)$.

Worked example

A solenoid valve, such as a dishwasher's or a digital shower's, shuts in an effective $t_v = 20$ ms at the end of $L = 8$ m of copper pipe, 10 mm outside diameter with a 0.6 mm wall. The flow is 6 L/min of water at 55 °C, and the rest pressure is 3 bar.

Quantity	Working	Result
Bore and area	$D = 10 - 2 \times 0.6 = 8.8$ mm; $A = \pi D^2/4$	6.08×10^{-5} m ²
Velocity	$v = Q/A = (6/60\,000)/A$	1.64 m/s
Water at 55 °C	$\rho = 985.6$ kg/m ³ , $c = 1546.8$ m/s, $K = \rho c^2$	2.36 GPa
Copper at 55 °C	$E = 115.9$ GPa, $\nu = 0.34$	
Restraint factor (anchored)	$2(0.6/8.8)(1.34) + 8.8(1 - 0.34^2)/9.4$	$\psi = 1.011$
Pipe term	$\psi KD/(Ee) = 1.011 \times 2.36 \times 8.8/(115.9 \times 0.6)$	0.302
Wave speed	$a = \sqrt{K/\rho} / \sqrt{1 + 0.302} = 1546.8/1.141$	1356 m/s
Critical time	$T_c = 2L/a = 16/1356$	11.8 ms ($< t_v$, slow)
Joukowsky	ρav	22.0 bar
Michaud	$2\rho Lv/t_v$	13.0 bar
Simulation	method of characteristics, $N = 20$ reaches	15.8 bar
Design surge Δp_d	larger of Michaud and simulation	15.8 bar $\gg$ 2 bar
Arrester volume	$\frac{1}{2}\rho ALv^2 = 0.65$ J; $p_0 = 4.01$ bar abs.; $n = 1.4$	$V_0 \approx 5$ ml
Slower valve	closure time giving a 2 bar surge	$t_v \geq 0.13$ s
Lower flow	$v \leq \Delta p_a/(\rho a)$	$Q \leq 0.55$ L/min

The design surge is far above the 2 bar limit, so mitigation is required. The peak pressure, $3 + 15.8 = 18.8$ bar, also exceeds the pipe rating, and the low-pressure wave that follows would break the column. Here an arrester at the appliance's valve is the practical answer.

What the screening shows

Figure 4 applies the model across the project's design range. Solenoid valves on small-bore pipe exceed the limit at almost any length once the flow passes 2 to 3 L/min. Lever mixers and quarter-turn taps are acceptable on short runs but fail on long small-bore ones. Reducing the bore to cut the volume of a dead leg raises the

Solenoid valves need protection on almost any small-bore run

Longest run in metres (valve to reflection point) needing no mitigation; 3 bar, 55 °C; capped at 30 m

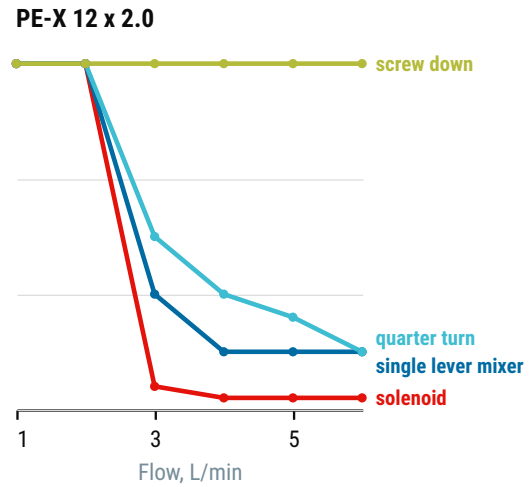
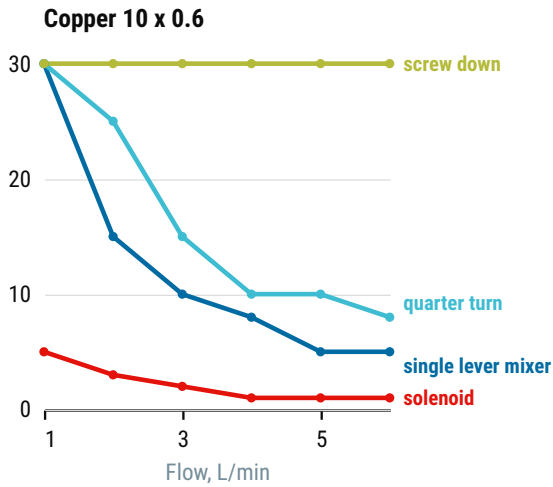


Figure 4: Screening results from the model for two common small-bore pipes. Each line is the longest run, from valve to reflection point, that needs no mitigation, at 3 bar and 55 °C. The full table, for six pipes, is in `results/waterhammer/screening.csv`.

velocity, and so the surge. The two design aims pull against each other, and this model is the check on that trade-off.

Limitations

- **One pipe, one valve.** Local losses at fittings, branches and unsteady friction are left out. A branched network needs a full transient analysis.¹⁴
- **Closure times.** The valve closure times are engineering estimates. Where a result is marginal, the valve should be measured.
- **Plastic moduli.** These are indicative, and multilayer pipe is treated as a single homogenised wall.
- **Column separation** is detected, not simulated.
- **Softening.** Trapped air and flexible connections soften real surges, so the results lean conservative.

¹⁴ Chaudhry (2014).

Symbols

Symbol	Meaning	Unit
A	bore cross-sectional area	m^2
a	wave speed	m/s
B	characteristic impedance, $a/(gA)$	s/m^2
c	speed of sound in water	m/s
C_P, C_M	characteristic constants	m
D	bore (inside diameter)	m
E	Young's modulus of the wall	Pa
e	wall thickness	m
f	Darcy friction factor	–
g	acceleration due to gravity	m/s^2
H, H_R, H_v	pressure head; at reflection point; at valve	m
h_f	friction head over the pipe	m
K	bulk modulus of water	Pa
k	valve coefficient, $(\tau Q_0)^2/H_{v0}$	m^5/s^2
L	length, valve to reflection point	m
N	number of reaches	–
n	polytropic index of the gas	–
p_s	rest pressure (gauge)	Pa
p_0	arrestor precharge (absolute)	Pa
p_{vap}	vapour pressure	Pa
Q, Q_0	flow rate; steady flow	m^3/s
R	friction coefficient per reach	s^2/m^5
Re	Reynolds number	–
T_c	critical time, $2L/a$	s
t, t_v	time; effective closure time	s
V_0	arrestor precharge volume	m^3
v	mean velocity	m/s
x	distance from the reflection point	m
$\Delta p, \Delta p_d, \Delta p_a$	surge; design surge; allowed rise	Pa
Δv	velocity change	m/s
ε	bore roughness	m
μ	dynamic viscosity	Pa s
ν	Poisson's ratio	–
ρ	water density	kg/m^3
τ	valve open fraction	–
ϕ, ψ	restraint terms	–

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The code, tests and data behind this note are in the GitHub repository `gwilma/deadleg_deadtime`. The file `docs/water_hammer.md` gives the limits and sources in brief, `docs/physics/figures.py` redraws every figure, and `excel/waterhammer.xlsx` implements the same model in a spreadsheet.